

Deep learning: Portfolio assignment 1

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In this portfolio assignment, several hyperparameters for neural networks were explored. This report reflects a small hypothesis experiment regarding the architecture of a neural network in relation to the learning process. Increasing the number of neurons in the layer(s) of a network increases model capacity, while increasing the number of epochs extends the training time. Both changes add complexity to the learning process. To see if increasing the number of epochs works for different layer sizes, the following hypothesis is tested:

H0: Increasing epochs affects validation performance equally for small and large networks.

H1: Increasing epochs has a different effect on validation performance for small versus large networks.

The experiment was set up with two epoch settings (3 and 10) and four neurons settings (64, 128, 256 and 512). All other hyperparameters remained fixed during the experiment.

Results

Units	Training loss		Validation loss	
	3 epochs	10 epochs	3 epochs	10 epochs
64	0.37	0.26	0.39	0.35
128	0.34	0.24	0.39	0.34
256	0.32	0.22	0.39	0.32
512	0.32	0.22	0.37	0.35

Increasing the number of epochs reduced training loss for all model sizes. However, the improvement in validation loss was not equal across different neurons settings. The validation loss between 3 and 10 epochs decreased at each step from 64 to 128 to 256 units. However, at 512 units the decrease in validation loss was marginal.

Also, the gap between training loss and validation loss was the highest at 512 units and 10 epochs, suggesting that a higher number of units and epochs tends to have an increasing risk of overfitting the data.

These results provide evidence against the null hypothesis that increasing the number of epochs has an equal effect for small and large networks.

